

IZMIR INSTITUTE OF TECHNOLOGY

Department of Computer Engineering

CENG463 - Introduction to Machine Learning

Take-Home Midterm Examination Question 4

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1 Introduction

Clustering is a fundamental task in unsupervised machine learning aimed at discovering latent structures and natural groupings within unlabeled data. While basic algorithms like K-Means are widely utilized due to their computational efficiency, they often fail to capture complex data geometries, such as non-spherical clusters, varying densities, or overlapping distributions. This study addresses these limitations by performing a comprehensive evaluation of four distinct clustering paradigms: centroid-based, distribution-based, density-based, and hierarchical methods.

The scope of this report encompasses the full machine learning lifecycle for clustering: from rigorous exploratory data analysis (EDA) and preprocessing to advanced model selection and stability testing. A key focus is placed on "Cluster Ensembles," where multiple base partitions are synthesized into a single consensus solution to enhance robustness and accuracy. By applying these methods to the Wholesale Customer [2] and Optdigits [3] datasets, we aim to demonstrate a deep understanding of algorithmic behavior, the impact of mathematical assumptions (e.g., Gaussianity, density connectivity), and the importance of both internal and external validation metrics in high-dimensional spaces.

2 Problem Formulation

The clustering task is formally defined as partitioning a dataset $X = \{x_1, x_2, \dots, x_n\}$ of n observations in a d -dimensional space $\mathbb{R}^d$ into K disjoint or overlapping subsets (clusters) $\{C_1, C_2, \dots, C_K\}$. The objective functions vary significantly across the algorithms implemented in this study:

- **Centroid-based (K-Means):** Formulated as a variance-minimization problem. The goal is to minimize the Within-Cluster Sum of Squares (WCSS), also known as inertia:

$$J = \sum_{i=1}^K \sum_{x \in C_i} \|x - \mu_i\|^2 \quad (1)$$

where μ_i is the mean of points in C_i .

- **Distribution-based (Gaussian Mixture Models):** Formulated as a density estimation problem. We assume data is generated from a mixture of K Gaussian distributions. The goal is to maximize the log-likelihood of the parameters $\theta = \{\pi_k, \mu_k, \Sigma_k\}$ using the Expectation-Maximization (EM) algorithm.
- **Density-based (DBSCAN):** Formulated as a reachability problem. Clusters are defined as maximal sets of density-connected points. A point p is a core point if at least *min_samples* are within distance ϵ of p .
- **Hierarchical (Agglomerative):** Formulated as a recursive merging process. Using Ward's linkage, we minimize the increase in total within-cluster variance at each merge step.

The "problem" is further complicated by the nature of the Wholesale Customer dataset. As shown in the Exploratory Data Analysis (Figure 1), the features exhibit heavy right-skewness and extreme outliers. This necessitates a log-transformation preprocessing step to satisfy the normality assumptions of models like GMM and to prevent K-Means from being biased by high-spending outliers. Furthermore, the high correlation between features (Figure 2) suggests that a simple Euclidean distance might not capture the true semantic separation between customer segments, requiring the use of PCA or UMAP for lower-dimensional interpretation.

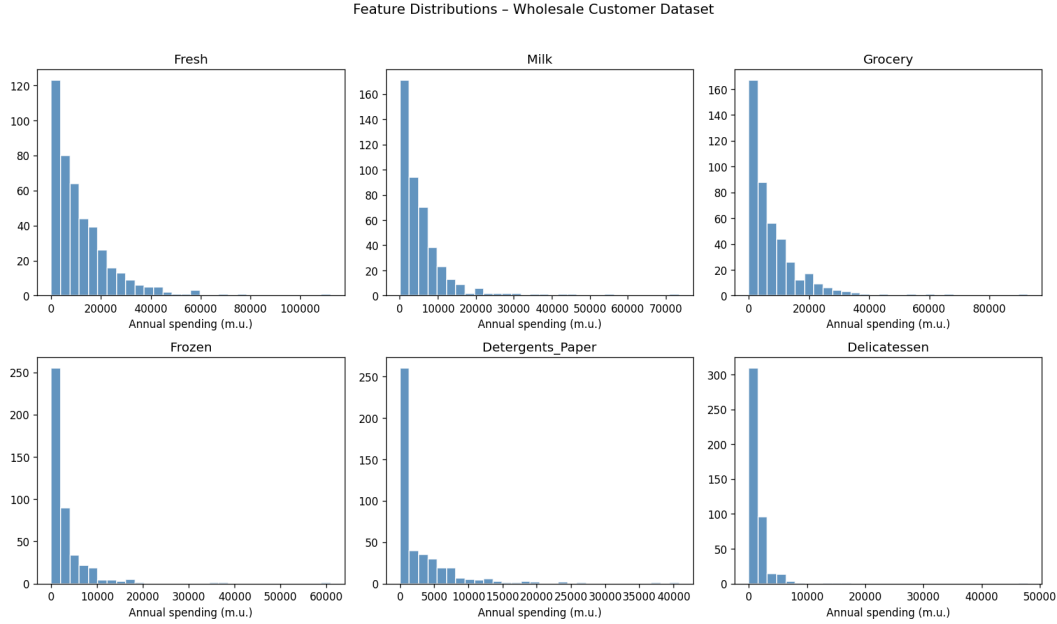


Figure 1: Histograms of the Wholesale Customer dataset categories. The extreme right-skewness observed across all features poses a significant challenge for distance-based clustering, necessitating a $\log(1 + x)$ transformation to ensure feature distributions are more Gaussian-like.

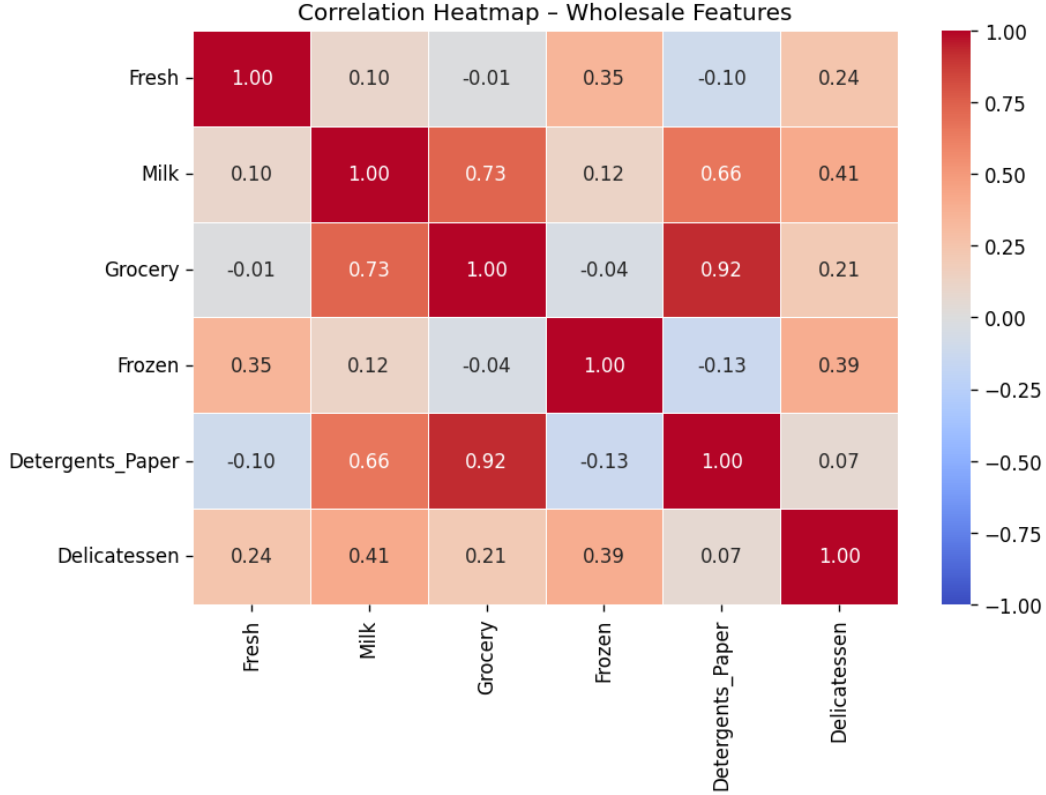


Figure 2: Correlation matrix of the features. The strong correlation (0.92) between Grocery and Detergents_Paper indicates redundant information, which can inflate the importance of certain spending patterns in clustering algorithms.

3 Experimental Setup

3.1 Dataset Characterization

The experimental evaluation is performed on two distinct datasets to test the algorithms under different structural and dimensionality conditions:

- **Wholesale Customer Dataset:** This dataset contains 440 observations representing annual spending of clients across six continuous product categories: *Fresh*, *Milk*, *Grocery*, *Frozen*, *Detergents_Paper*, and *Delicatessen*. It is utilized to identify latent customer segments in an unsupervised setting where ground-truth labels are unavailable.
- **Optdigits (Handwritten Digits):** Used for external validation, this dataset consists of 1,797 samples of handwritten digits (0-9). Each sample is an 8×8 grayscale image flattened into a 64-dimensional feature vector. The presence of true labels allows for the calculation of objective performance indices.

3.2 Preprocessing and Implementation Pipeline

To satisfy the mathematical assumptions of the various clustering paradigms and ensure reproducibility, a standardized preprocessing pipeline was established:

1. **Log Transformation:** As identified in the EDA phase, the features of the Wholesale dataset exhibit significant right-skewness. To address this, a log-transformation using $x' = \ln(1 + x)$ was applied, which suppresses the magnitude of extreme outliers and stabilizes the variance across categories.
2. **Feature Scaling:** Following transformation, the data was scaled using `StandardScaler` to achieve zero mean and unit variance. This is a critical step for centroid-based (K-Means) and distribution-based (GMM) algorithms, which are sensitive to the absolute magnitude and variance of features.
3. **Manifold Learning and Visualization:** For 2D interpretation of clusters, Principal Component Analysis (PCA) and Uniform Manifold Approximation and Projection (UMAP) were utilized to project the high-dimensional data into a visually interpretable space.

3.3 Evaluation Metrics

The study employs a dual evaluation strategy consisting of internal and external validation frameworks.

3.3.1 Internal Validation Metrics

For datasets without ground truth (Wholesale), the following metrics are used to assess the spatial quality of the clusters:

- **Silhouette Score:** Measures how similar an object is to its own cluster compared to neighboring clusters, ranging from -1 to 1.
- **Calinski-Harabasz (CH) Index:** The ratio of the sum of between-cluster dispersion and within-cluster dispersion; higher values indicate better-defined clusters.
- **Davies-Bouldin (DB) Index:** Represents the average similarity ratio of each cluster with its most similar cluster, where lower values signify superior separation.

3.3.2 External Validation Metrics

For the Optdigits dataset, clustering performance is benchmarked against ground-truth labels using:

- **Adjusted Rand Index (ARI):** Measures the similarity between true and predicted labels while adjusting for chance grouping.
- **Normalized Mutual Information (NMI):** A normalization of the Mutual Information score that scales results between 0 and 1.

- **Fowlkes-Mallows Index (FMI):** The geometric mean of the pairwise precision and recall.

3.4 Software and Environment

All experiments were implemented in Python 3 using the `scikit-learn` library for core clustering algorithms. Supplemental libraries include `NumPy` and `Pandas` for data manipulation, `SciPy` for hierarchical linkage calculations, and `Matplotlib/Seaborn` for visualization. To ensure the reproducibility required by the midterm guidelines, a global random seed of 42 was fixed throughout all stochastic operations.

4 Methodology and Parameter Selection

The effectiveness of clustering is highly dependent on the selection of appropriate hyperparameters. This section details the systematic approach used to determine the optimal parameters for each of the four algorithms implemented.

4.1 K-Means Clustering

K-Means was evaluated across a range of $k \in [2, 10]$ using three distinct criteria: the Elbow method (Inertia), Silhouette Score, and the Gap Statistic. As illustrated in Figure 3, the Silhouette Score peaks at $k = 2$, indicating the most distinct separation between clusters. However, the Gap Statistic, which compares the change in within-cluster dispersion to a null reference distribution, identifies $k = 4$ as the optimal number of clusters. For the final implementation on the Wholesale dataset, $k = 2$ was selected to maximize the Silhouette coefficient (0.2903).

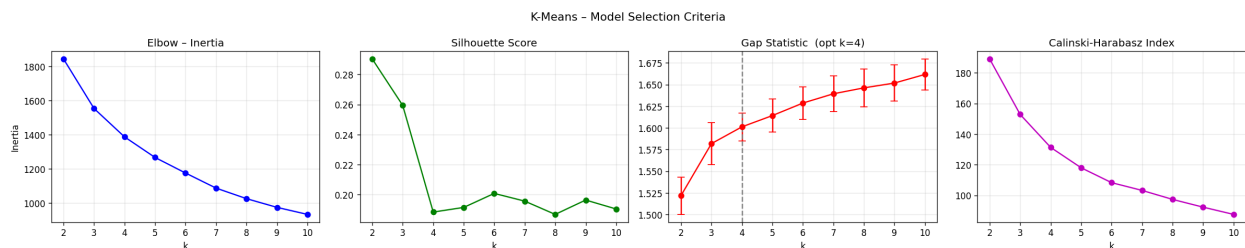


Figure 3: K-Means model selection plots: Inertia (elbow), Silhouette Score, Gap Statistic (identifying $k = 4$), and Calinski-Harabasz Index.

4.2 Gaussian Mixture Model (GMM)

Unlike the hard assignments of K-Means, GMM employs an Expectation-Maximization (EM) approach to provide probabilistic cluster memberships. Model selection was conducted using the Bayesian Information Criterion (BIC) and the Akaike Information Criterion (AIC).

As shown in Figure 4, the BIC curve reaches its minimum at 4 components, suggesting the best balance between model fit and complexity. The AIC curve, which penalizes complexity less severely, suggests a higher component count of 8. Following the principle of parsimony, 4 components were chosen for the GMM analysis.

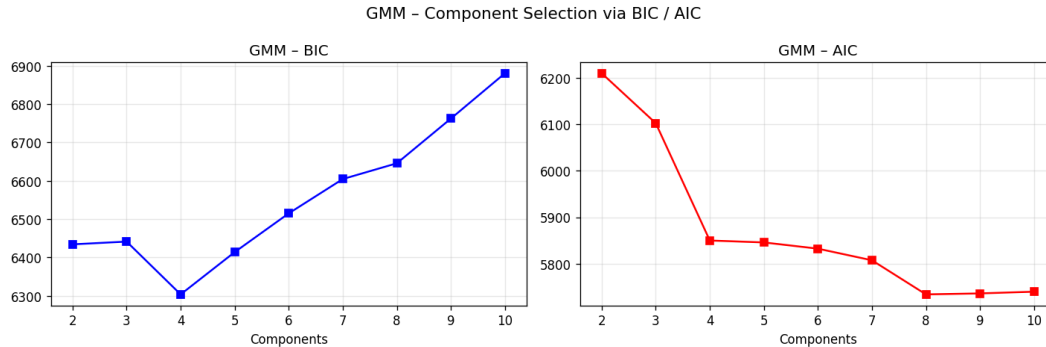


Figure 4: GMM component selection via BIC and AIC. The BIC suggests a 4-component mixture as the most appropriate model.

4.3 DBSCAN

DBSCAN requires the tuning of two critical parameters: the neighborhood radius (ϵ) and the minimum number of points required to form a dense region (*min_samples*). To find a suitable ϵ , a k -distance graph was generated with $k = 11$ (calculated as $2 \times \text{dim} - 1$). As seen in Figure 5, the "knee" of the curve, representing the point where density begins to drop significantly, occurs between 1.0 and 1.5. A subsequent grid search over ϵ values and *min_samples* identified $\epsilon = 1.0006$ and *min_samples*=7 as the configuration yielding the highest Silhouette score.

4.4 Agglomerative Clustering

Hierarchical clustering was implemented using Ward's linkage, which minimizes the total within-cluster variance. A dendrogram was generated on a subsample of 200 points to visualize the merge hierarchy. In Figure 6, the large vertical distance between the final two branches confirms that a two-cluster solution is the most natural partition for this data. This visual evidence is supported by the Silhouette analysis, which also identified $k = 2$ as the best-performing cluster count for the Agglomerative method.

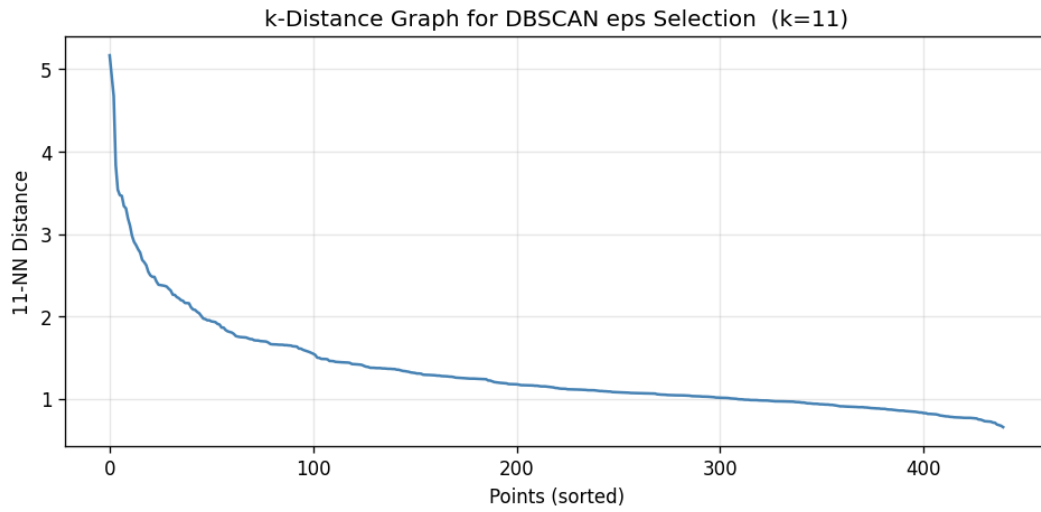


Figure 5: k-Distance graph for DBSCAN parameter selection ($k = 11$). The elbow point suggests an ϵ near 1.0.

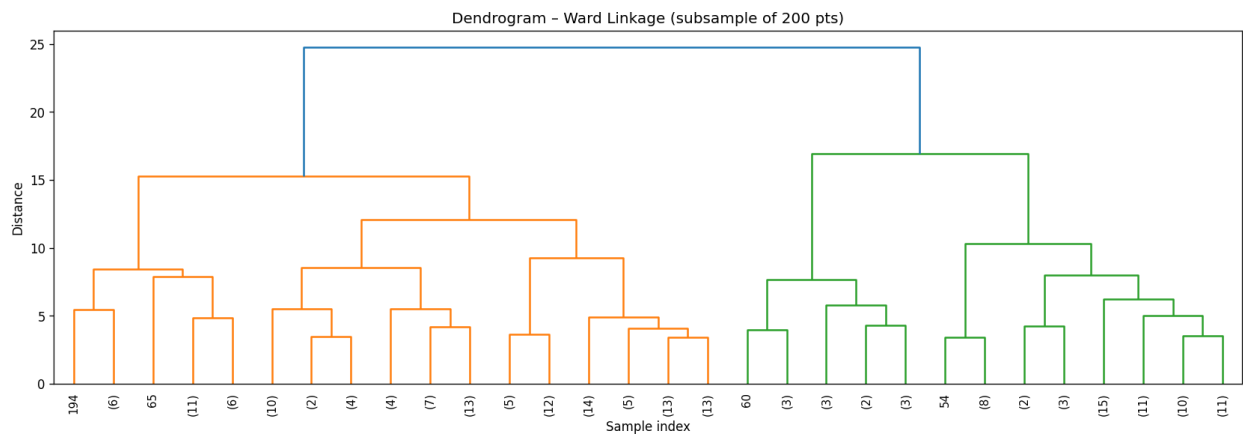


Figure 6: Dendrogram using Ward linkage. The clear separation into two primary branches justifies a two-cluster approach.

5 Evaluation and Results

This section presents a quantitative evaluation of the clustering performance across both datasets. We utilize internal metrics for the unlabeled Wholesale Customer dataset and external metrics for the Optdigits dataset to provide a comprehensive assessment of algorithmic efficacy.

5.1 Internal Validation (Wholesale Customer Dataset)

The performance on the Wholesale Customer dataset was evaluated using Silhouette Score, Calinski-Harabasz (CH) Index, and Davies-Bouldin (DB) Index. The results are summarized in Table 1.

Table 1: Internal Clustering Metrics for Wholesale Customer Dataset

Model	Clusters	Silhouette	CH-Index	DB-Index
K-Means	2	0.2903	189.05	1.3515
GMM	4	0.1717	75.73	2.3411
DBSCAN	2	0.0608	14.13	3.7404
Agglomerative	2	0.2585	134.62	1.6004
Ensemble	2	0.2896	188.52	1.3546

The internal metrics indicate that K-Means achieved the highest Silhouette score (0.2903) and CH-Index (189.05), while maintaining the lowest DB-Index (1.3515), identifying it as the strongest individual performer for this dataset. Interestingly, the Ensemble method closely matched K-Means performance (0.2896 Silhouette), suggesting that the consensus approach successfully stabilized the groupings without sacrificing internal cohesion. DBSCAN performed poorly in this high-dimensional space, yielding a Silhouette score near zero (0.0608), which suggests that the density-based approach struggled to separate clusters in the presence of noise or overlapping distributions.

5.2 External Validation (Optdigits Dataset)

To verify the algorithms’ ability to recover true semantic structures, we applied them to the Optdigits dataset. Performance was measured against ground-truth digit labels using Adjusted Rand Index (ARI), Normalized Mutual Information (NMI), and Fowlkes-Mallows Index (FMI).

On the Optdigits dataset, Agglomerative Clustering significantly outperformed all other models, achieving an ARI of 0.6643 and an NMI of 0.7956. This indicates that the hierarchical relationships and Ward’s linkage criterion are particularly well-suited for capturing the structural variations in handwritten digit patterns. While K-Means remained competitive, GMM and DBSCAN showed lower alignment with the true labels, likely due to the complexity of the 64-dimensional feature space where Gaussian assumptions or global density thresholds are less effective.

Table 2: External Validation Metrics for Optdigits Dataset (10 True Classes)

Model	Clusters	ARI	NMI	FMI
K-Means	10	0.4780	0.6319	0.5434
GMM	10	0.3604	0.6083	0.4671
Agglomerative	10	0.6643	0.7956	0.7111
DBSCAN	12	0.1822	0.4661	0.3579

6 Stability and Ensemble Analysis

To ensure the robustness of our clustering results, we performed a stability analysis using bootstrap resampling and implemented a cluster ensemble method to achieve a consensus partition.

6.1 Cluster Stability Analysis

The stability of each algorithm was evaluated by subsampling 80% of the Wholesale Customer dataset over 20 independent runs. For each run, the Adjusted Rand Index (ARI) was computed between partitions to measure the consistency of the cluster assignments.

As shown in Figure 7, K-Means demonstrated the highest stability with a mean ARI of 0.9560 ± 0.0514 . GMM followed with a mean ARI of 0.8149 ± 0.0754 , indicating relatively stable probabilistic assignments. In contrast, Agglomerative Clustering exhibited the lowest stability (0.5115 ± 0.1874), suggesting that its hierarchical merging process is highly sensitive to the initial sample composition. DBSCAN maintained moderate stability with an ARI of 0.7545 ± 0.0667 .

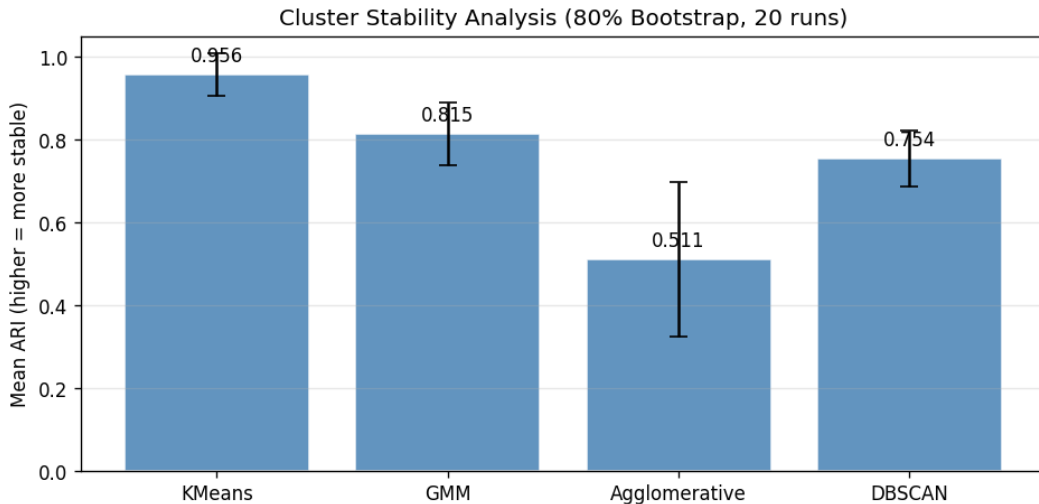


Figure 7: Cluster Stability Analysis (80% Bootstrap, 20 runs). K-Means emerges as the most consistent algorithm across different data subsets.

6.2 Cluster Ensemble Implementation

We implemented a cluster ensemble using a Co-Association Matrix to combine the strengths of various models. The ensemble integrated 13 base partitions, including K-Means (with varying k), GMM, and DBSCAN. The final consensus clustering was derived by applying Agglomerative Clustering to the dissimilarity matrix obtained from the co-association values.

Figure 8 visualizes the co-association matrix for a subsample of 150 points. The two distinct, bright blocks along the diagonal indicate a high probability of points being co-clustered across multiple base models, signifying a strong consensus for a two-cluster structure.

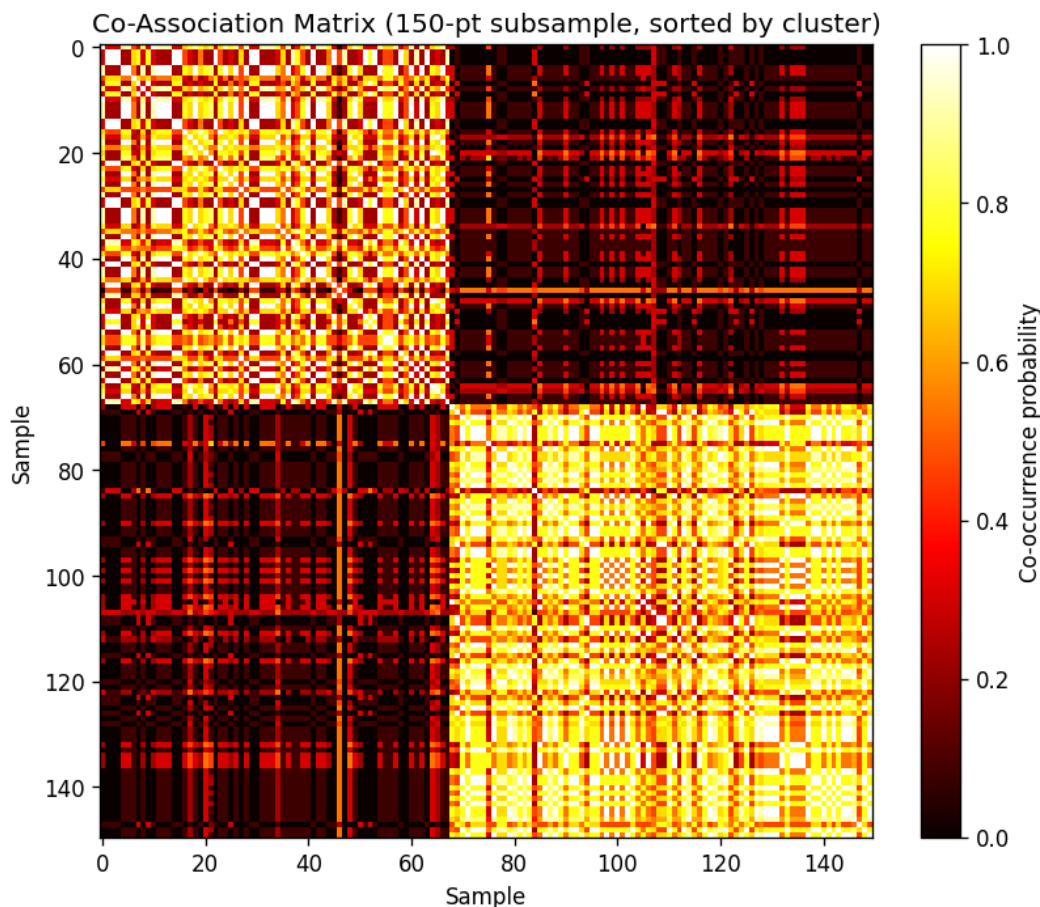


Figure 8: Co-Association Matrix Heatmap (sorted by cluster). The clear block-diagonal structure confirms the stability of the two-cluster consensus solution.

Quantitatively, the ensemble achieved a Silhouette score of 0.2896, nearly identical to the performance of the best-tuned K-Means (0.2903). This demonstrates that the ensemble method successfully captured the core structure of the data while mitigating the biases of individual algorithms.

7 Visualization and Discussion

This section provides a qualitative analysis of the clustering results through dimensionality reduction and a discussion on the theoretical assumptions underlying each algorithm.

7.1 Visual Analysis of Wholesale Clusters

To interpret the high-dimensional clusters in the Wholesale Customer dataset, we utilized Principal Component Analysis (PCA) and Uniform Manifold Approximation and Projection (UMAP).

As shown in Figure 9, the first two principal components account for 71.3% of the total variance (PC1: 44.1%, PC2: 27.2%). K-Means and Agglomerative Clustering produce very similar, well-separated partitions along the PC1 axis, which corresponds to the overall scale of spending. GMM (with $k = 4$) identifies more granular sub-segments, although there is visible overlap between the Gaussian components in the 2D projection. DBSCAN identifies a large central cluster (C0) but classifies a significant portion of the data as noise (shown in dark blue), highlighting its sensitivity to outliers even after log-transformation.

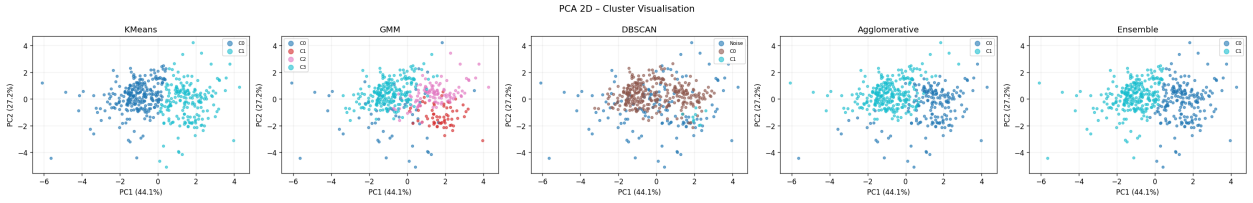


Figure 9: PCA 2D projection of clustering results on the Wholesale Customer dataset. K-Means and Agglomerative methods show the cleanest separation.

The UMAP projection in Figure 10 confirms these observations. UMAP, which preserves local manifold structure better than PCA, shows that the two-cluster solution (K-Means/Agglomerative/Ensemble) aligns with two dense regions of the manifold. The ensemble method successfully captures this global structure.

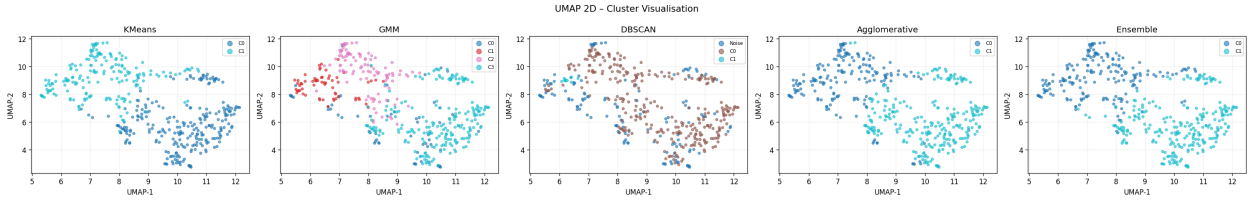


Figure 10: UMAP 2D projection. The non-linear embedding emphasizes the distinctness of the two-cluster solution chosen by the ensemble.

7.2 Visual Analysis of Optdigits Validation

For the Optdigits dataset, we visualized the ground-truth labels against the K-Means predictions. Figure 11 illustrates that while K-Means (ARI: 0.4780) recovers the general location of most digit classes, there is significant overlap in the center of the PCA space. Agglomerative clustering, which achieved the highest ARI (0.6643), likely outperformed K-Means by capturing the hierarchical stroke-based similarities between digits that simple Euclidean centroids miss.

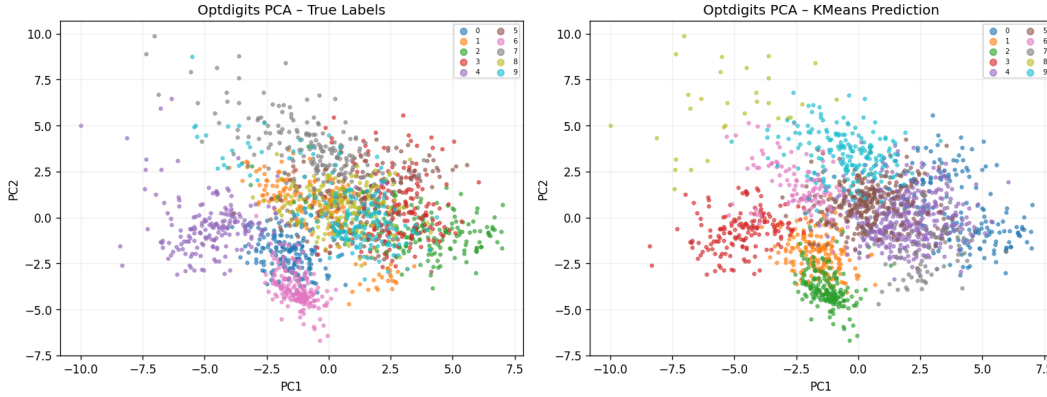


Figure 11: Optdigits PCA visualization comparing true labels (left) with K-Means predictions (right). Most classes are recovered, though some overlap persists in the 2D projection.

7.3 Discussion of Algorithmic Assumptions

The performance variations observed across the algorithms can be attributed to their fundamental assumptions:

- **K-Means:** Assumes spherical clusters of similar size and density. By log-scaling the data, we effectively made the clusters more spherical, allowing K-Means to achieve high stability and silhouette scores.
- **GMM:** Assumes the data is a mixture of Gaussian distributions with arbitrary ellipsoidal shapes. While more flexible than K-Means, the BIC selection of $k = 4$ suggests that the "true" clusters might contain nested sub-groups, though these are less stable under bootstrap resampling.
- **DBSCAN:** Assumes clusters are continuous regions of high density. The high noise count in the Wholesale dataset indicates that many customers have unique spending patterns that do not belong to a dense neighborhood, making density-based methods less effective for this specific task.
- **Agglomerative (Ward):** Assumes a hierarchical structure. While excellent for digits (which have sub-patterns), it showed low stability on the Wholesale data, likely because the specific spending outliers can trigger different branch merges early in the process.

8 Conclusion

The experimental results conclude that for the Wholesale Customer dataset, a two-cluster solution is the most robust and stable, effectively distinguishing between different scales of retail and hotel/restaurant spending. While K-Means provided the most stable individual results, the Cluster Ensemble via a co-association matrix offered a powerful consensus that balanced the biases of multiple algorithms. For the more complex Optdigits dataset, Agglomerative Clustering proved superior, demonstrating that hierarchical methods are better suited for data with intricate, non-linear class boundaries.

Repository and Reproducibility

GitHub Link: <https://github.com/behicekadioglu/CENG463---Take-Home-Midterm/tree/main>

References

- [1] Prof. Dr. Aytuğ Onan, *CENG 463 Machine Learning Take Home Midterm*, Izmir Institute of Technology, 2026.
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